



AI and Democracy: Mapping Notions, Debates and Challenges

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Abstract

Artificial intelligence (AI) is widely discussed as either a threat to democracy or a means of strengthening it. But democracy, an essentially contested concept, invokes many different levels at which AI may assist and obstruct the legitimacy of democratic government. Therefore, the aim of this review is to map how the democratic value of AI, whether positive or negative, depends on different democratic models. The review examines what there is currently no examination of: how democracy is conceptualized, what the cross-cutting patterns are, and how we should prescriptively use democracy in relation to assessing how AI supports or undermines democracy. From this, the review maps three dynamics currently at play in the literature. It shows that conceptual unclarity and pessimistic views on AI's contribution to democracy often coincide, prescribing the need for conceptual clarity. It shows that democratic models are often “inherited” rather than reworked, prompting the need for greater engagement with how AI reshapes democratic theory. And it maps how democratic assumptions shape evaluations of AI's democratic implications. From these results, we assess the limits of existing scholarship: without greater conceptual clarity regarding democracy, empirical findings and normative arguments risk talking past one another. Taken together, this review suggests that progress in research on democracy and AI depends not only on further empirical investigation or technical refinement, but also on sustained conceptual work that makes democratic assumptions explicit and thus open to scrutiny.

Keywords Democracy · Artificial intelligence · Political philosophy · Ethics · Legitimacy

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1 Introduction

Artificial intelligence (AI) is frequently framed as either a threat to democracy or a means of revitalizing it. Concerns about algorithmic misinformation, manipulation, and surveillance are often contrasted with promises of enhanced participation, improved public administration, and more responsive representation. Across these debates, “democracy” serves as a central normative benchmark for evaluating the political implications of AI. Yet despite its prominence, it is often unclear what conception of democracy is being invoked. Mapping these invoked conceptions of democracy is the focus of this review.

Existing reviews have mapped citizens’ attitudes toward AI (del Álamo Cienfuegos et al., 2024; Isoieva et al., 2024; Pham et al., 2025; Jensen & Chen, 2025; Howell, 2025), surveyed national AI strategies (Fatima et al., 2021), and explored relationships between democratic institutions and AI development (Chehoudi, 2025). However, far less attention has been paid to how democracy itself is conceptualized when scholars explicitly assess the relationship between AI and democratic governance.

This omission is consequential because democracy is an essentially contested concept (Gallie, 1955). Its meaning is inseparable from normative disagreement and competing theoretical traditions. Rather than referring to a single institutional arrangement, value, or norm, democracy encompasses multiple conceptions that prioritize different, and sometimes conflicting, principles or dimensions, such as participation, representation, or deliberation. As a result, assessments of whether AI strengthens or undermines democracy depend on which conception of democracy is being employed.

This scoping review maps how democracy is conceptualized in scholarship on AI and analyses how these conceptualizations shape assessments of AI’s implications for democratic governance. It examines how democracy is discussed in relation to democratic theory and how these conceptualizations influence whether AI is framed as strengthening or undermining democratic practices. By identifying cross-cutting patterns in how democracy is invoked, the purpose of the review is to provide a clearer conceptual foundation and starting point for debates about AI and democratic governance.

Accordingly, the review addresses the following research questions:

- RQ1: What notions of democracy are discussed in relation to AI?
- RQ2: Which cross-cutting democratic debates and dynamics are most prevalent in this literature, and how are they linked to assessments of AI?

Answering these questions allows the review to make three analytical contributions. First, it demonstrates that conceptual ambiguity about democracy often coincides with pessimistic assessments of AI’s democratic implications, highlighting the need for greater conceptual clarity. Second, it shows that many studies rely on “inherited” assumptions about democracy rather than reworking them when assessing AI, underscoring the need for greater engagement with how AI reshapes democratic theory. Third, it maps how democratic *assumptions* shape evaluations of AI’s democratic *implications*.

The review ends by assessing the limits of existing scholarship: without greater conceptual clarity regarding democracy, empirical findings and normative arguments risk talking past one another. This means that progress in research on democracy and AI depends not only on further empirical investigation or technical refinement, but also on sustained conceptual work that makes democratic assumptions explicit and open to scrutiny.

2 Background

When values such as transparency, responsibility, equality, and accountability are examined in relation to AI, they are frequently framed as contributing to societal progress toward democracy. In this sense, democracy often functions as an implicit normative horizon within debates on AI ethics and governance. While this literature is conceptually rich, and has itself been the subject of reviews mapping ethical positions and frameworks (Mittelstadt et al., 2016; Tsamados et al., 2022), the explicit relationship between AI and discussions of democracy has not, to our knowledge, been the subject of a structuring review.

Democracy is routinely invoked as a legitimizing aim for AI development and use. However, the underlying logics of this normative framework are rarely analysed in detail, although exceptions may be found like that of Mathias Risse's (2023) work. References to democratic values, democratic discourse, democratic ideals, democratic processes, and democratic institutions frequently appear without explicit theoretical elaboration or discussion (Kay et al., 2024; Yun et al., 2024; Agostino et al., 2025; Presno Linera & Meuwese, 2025; Vinay et al., 2025; Ariel & Elishar, 2025; Paraschou et al., 2025). Examples of thin or overly general invocations of democracy (which, to be sure, does not mean that these invocations are necessarily mistaken) include claims that AI policy dissemination contributes to "the broader goal of enhancing democratic engagement and accessibility in the digital age" (Yun et al., 2024, p. 16) or assertions that "[b]iased LLMs' decisions and misinformation undermine democratic processes" without much further theoretical explication (Paraschou et al., 2025, p. 1715).

Similar patterns can be observed in adjacent research domains, including studies of democratic decline or democratic potential in education (Macgilchrist et al., 2020; Gray, 2020; Thompson et al., 2023; Aleman et al., 2024; Bulathwela et al., 2024; Peters & Green, 2024; Wieczorek, 2025), analyses of misinformation, deepfakes, and their democratic consequences (Vaccari & Chadwick, 2020; Ascott, 2020; Kopecky, 2024; Sanchez-Acedo et al., 2024; Lundberg & Mozelius, 2025; Bhandari & Bhandari, 2025), and work on technological strategies for detecting or mitigating such threats (Wilder & Vorobeychik, 2019; Kaushal et al., 2022; Rajalaxmi et al., 2023; Lal & Saini, 2023; Berjawi et al., 2023; Dan, 2025). Further examples include research on voting systems and digital governance tools (Plüss et al., 2018; Azhaguramyaa et al., 2025; Srinivas et al., 2025), labour markets (McGaughey, 2022), and free speech and the public sphere (Alkiviadou, 2022; Acosta Navas, 2025). Across these diverse domains, democracy functions less as an analytically well-defined notion than as a broadly mobilized normative reference.

While this review is the first to investigate the conceptual clarity of notions of democracy invoked in debates on AI, there are other reviews and surveys on democracy and

AI with different foci. The great majority of reviews about democracy and AI focus on empirical works studying either citizen attitudes or democratic development: Djen et al. (2023) reviews which areas of AI research may support government-building toward democracy, while Pham et al. (2025) survey European citizens' beliefs about AI's impact on democracy. Fatima et al. (2021) examine 34 national AI plans, finding that highly democratic countries are less likely than lower-scoring ones to favour ethical and governmental concerns related to AI. Chehoudi (2025) finds a negative correlation between democracy and AI development, concluding that "higher AI development is associated with lower levels of democracy". Studying societal attitudes, del Álamo Cienfuegos et al. (2024) show that democracy is the domain most associated with fears of negative impact, in contrast to the health domain, which is linked to optimism. Similarly, Jensen and Chen (2025) survey positive and negative views of AI across class divisions and attitudes toward governments, while Howell (2025) investigates perceptions of AI in democratic and non-democratic countries. Isoieva et al. (2024) focus on young people's concerns about AI's impact on society. The ninth and less directly empirical study that we are currently aware of is Zaytsev and Kuskova (2025) who code the AI-and-democracy literature thematically and compare it with media framings, identifying a gap between academia and public discourse. Together, these studies highlight growing concern about AI's relationship to democracy. They do not, however, provide a systematic overview of how democracy itself is understood and discussed in the literature on AI and democracy.

Our analytical focus is on contributions in which authors explicitly invoke the label "artificial intelligence" to articulate a relationship with democracy, rather than on studies that exclusively address more specific descriptors from computer science. Conceptually, this approach draws on Sheila Jasanoff's (2015, p. 4) account of sociotechnical imaginaries as "collectively held, institutionally stabilized, and publicly performed visions of desirable futures." We adopt this framework in a weaker, analytically cautious sense, treating sociotechnical imaginaries not as fully stabilized or consensual formations but as provisional and contested reference points. In this sense, AI approximates a sociotechnical imaginary: it may lack institutional consolidation and normative agreement, but it nonetheless operates as a focal term through which possible (desirable and undesirable) futures are articulated. Treating AI as an approximate sociotechnical imaginary enables us to examine how contemporary scholarship mobilizes the label "AI" to negotiate the relationship between emerging computational technologies and democratic governance. Accordingly, our review does not seek to evaluate whether particular technologies should properly be classified as AI, but examines how the term "AI" functions within scholarly discourse as a reference point through which democratic futures, opportunities, risks, and institutional responses are conceived and debated.

3 Method

We chose a scoping review to allow broader mapping of the literature revolving around AI and democracy (Munn et al., 2018). To ensure stringency and transparency we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines (Tricco et al., 2018).

3.1 Search Protocol

To allow the emergence of a wide range of topics and research fields working with AI and democracy we chose a short search string focusing on our main concepts. To ensure that democracy was a central concept in the papers we limited the search string to use “democracy”, without including related words such as the public sphere, totalitarianism, rule of law, human rights etc. Due to our focus on AI as a sociotechnical imaginary, we limited our search string solely to “artificial intelligence” instead of including related concepts such as algorithms, big data and machine learning. We chose to include the variation “democratic” but not “democratization”, as the latter concept often pertains to a distinct discussion, typically referring to AI accessibility or AI literacy (see Subramonian et al., 2024). In our assessment, articles addressing democratization in connection with democracy-related notions or debates would still be captured through our keywords “democracy” or “democratic”. These considerations resulted in the search string: *(ai OR “artificial intelligence”) AND (democracy OR democratic)* for paper title, abstract and keywords in the databases Scopus, Web of Science and PhilPapers extracted in July 2025 and updated in January 2026 to include all publications up to and including 2025. No lower time limit was set for the search.

3.2 Eligibility Criteria

First and foremost, we limited our pool of papers to peer-reviewed journal articles and conference proceedings in English. We also excluded any papers without AI as their main focus. Examples of this were papers centred on digital technologies in general, including AI but not with AI as their primary focus. Furthermore, we excluded papers where it was not clear whether their technological focus included AI. Examples of this were papers about algorithms or social media without explicitly pointing out whether these were considered *because* they are AI driven technologies or analysed as technologies without special interest in their AI components.

Finally, one or more notions of democracy needed to be discussed in relation to AI in the included papers, as we examine the possible democratic shifts happening due to AI. Therefore, we excluded papers that only shortly mention democracy, had democracy as a theme without further explication, applied a set understanding of democracy without discussion of the overall theory or its premises, or saw democracy as an existing presumption for further discussion. Instead, we included papers (i) discussing multiple notions of democracy in relation to AI, (ii) papers explicitly discussing how theoretical premises for democracy are affected by AI, and (iii) papers considering new foundations or types of democracy due to AI. These dimensions also needed to be made explicit by the authors, instead of implicitly assumed.

3.3 Screening and Coding Process

We used Covidence to facilitate the screening process, first a title and abstract screening followed by a full text screening. JW and AJ independently screened all papers and resolved conflicts through discussion resulting in a unanimous decision. In cases

of doubt, the screened article was always passed on (“included”) to the next phase to ensure further and more elaborate review. The full text exclusion categories are not mutually exclusive, e.g. a paper can be out of scope for this review both due to their lack of engagement with democracy and main focus on non-AI technologies (see Fig. 1).

For the included full texts an extraction template was developed to map relevant data and ensure systematic data extraction. During reading, we extracted: (i) notions of democracy, (ii) definition of democracy (if provided), (iii) types of AI noted or discussed, and (iv) topic domains. In addition, we documented (v) whether AI was framed as threatening or supporting democracy, and we logged how democracy was discussed by summing up in prose and highlighting key passages. Following data extraction JW and AJ compared notes and settled disagreements unanimously, creating the final dataset (see Table 4, in Appendix). This was done to ensure consistency and accurate data analysis.

4 Findings and Analysis

4.1 Background

4.1.1 Journals

The journals most prevalent in the corpus are *AI and Society*, which functions as the primary hub of this discussion ($n=11$), followed by *Philosophy & Technology* ($n=5$) and *Technology in Society* ($n=3$). Only *Minds and Machines* and *Science and Engineering Ethics* reach $n=2$ each, while the remaining publications are dispersed across 34 different journals and conferences. This long tail indicates that scholarship on AI and democracy is widely distributed, with only a small number of outlets playing a central role. Notably, journals oriented toward political philosophy are weakly represented. Within our sample, ten journals containing *political science*, *democracy*, *policy*, or *government* in their titles appear only once each ($n=1$), indicating a dispersed rather than concentrated presence of core political science venues. Computer science-adjacent venues such as *FAccT*, *AIES*, *CEUR*, and *Proceedings of Machine Learning Research* appear sporadically, a pattern consistent with the primarily conceptual and interpretive orientation of the literature. Overall, this distribution suggests that AI–democracy debates are articulated mainly through interdisciplinary and ethics- or philosophy-oriented venues rather than through disciplinary political theory journals.

4.1.2 Publication Year

Discussion of AI and democracy is largely absent in the period 2015 ($n=1$), 2016 ($n=0$) and 2017 ($n=1$), before increasing modestly in 2020 ($n=3$) and 2021 ($n=3$) (see Fig. 2). Publication volume rises sharply in 2022 ($n=11$), a peak that cannot be attributed to the public release of ChatGPT on 30 November 2022, which occurred too late to have influenced publication output in that year. Instead, six of the eleven

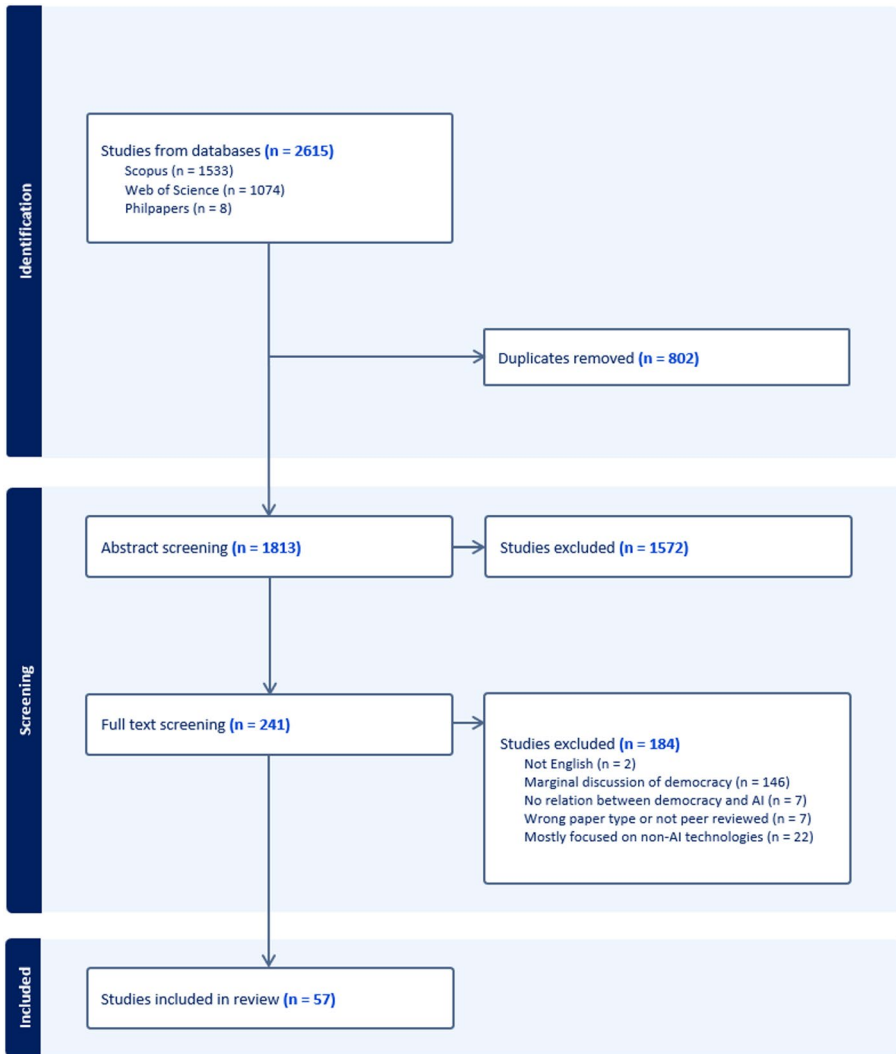


Fig. 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) flow chart

publications from 2022 focus explicitly on governance (Gianni et al., 2022; Erman & Furendal, 2022), political decision-making (König & Wenzelburger, 2022; Cavaliere & Romeo, 2022) or EU AI policy or national AI strategy (Carlsson & Rönblom, 2022; Paltieli, 2022). This concentration suggests that the sudden increase may be associated with policy developments following the European Commission's proposal to regulate AI in April 2021, later culminating in the AI Act. Contrary to our expectations that regulatory activity and the widespread release of conversational AI systems would immediately stimulate scholarly output, publication volume declines in 2023 ($n=7$) and remains comparatively stable in 2024 ($n=10$). A marked, dou-

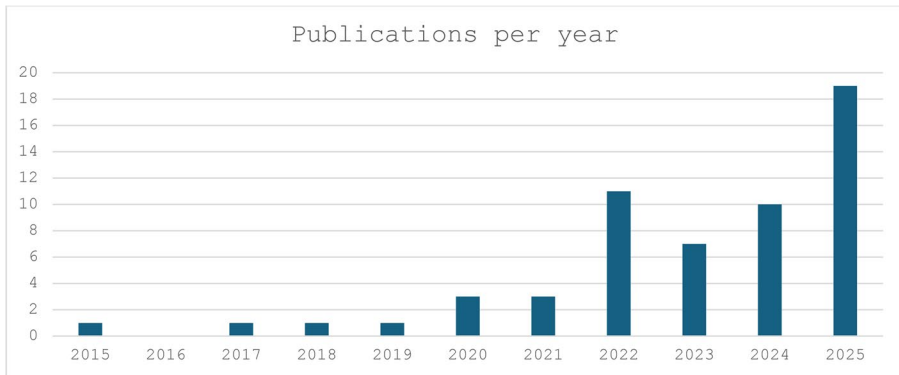


Fig. 2 Number of publications on AI and democracy per year

bling increase is observed in 2025 ($n=20$), indicating a significant growth in scholarly attention to questions of democracy in relation to AI. While publication patterns beyond the scope of the data cannot be assessed robustly, the overall trajectory points toward increasing engagement with democratic concepts over time.

4.1.3 Types of AI

Across the literature, the type of AI addressed remains under-specified in 39% of the articles ($n=22$), while 61% ($n=35$) explicitly refer to one or more specific types of AI. The categories presented in Table 1 are not mutually exclusive, as individual studies often mention multiple forms of AI; for example, machine learning and large language models (LLMs) may be discussed within the same contribution. Among the articles that specify a type, LLMs (18%) and machine learning (ML) (19%) are the most frequently referenced. By contrast, only 11% of the total sample ($n=57$) refers to superintelligence, a comparatively low proportion given the prominence of this form of AI in public debates.

Type specification does not, however, necessarily coincide with definitional clarity. 30 of the 57 articles (53%) do not provide an explicit definition of AI. Notably, 13 articles refer to particular types of AI without defining AI as a broader category. This pattern suggests that technical labels such as ML or LLMs often function as shorthand references in place of analytically grounded definitions, contributing to the broader conceptual unclarity observed across the literature.

Table 1 Types of AI mentioned in the reviewed literature (categories are not mutually exclusive)

Types of AI (not mutually exclusive)	
Large Language Models	10
Machine Learning	11
Superintelligence	6
Narrow AI	5
Artificial General Intelligence	5
Deep Learning	3
Other	14

Moreover, the literature exhibits substantial overlap and inconsistency in how different levels of abstraction within AI are articulated, making it difficult to assess the analytical level at which AI has democratic implications. In particular, categories such as machine learning (ML), deep learning (DL), and large language models (LLMs) are not deployed in a congruent or hierarchical manner. From a technical perspective, ML is commonly understood as a broad class of computational approaches, within which DL constitutes a subset, and LLMs represent a specific application class enabled by DL architectures. In the literature, however, these relationships are not consistently reflected. Some articles treat AI, ML, DL, and LLMs as interchangeable terms, while others juxtapose them with orthogonal classificatory schemes, such as narrow versus broad (or general) AI, which are categories within which both ML- and DL-based systems typically fall. As a result, multiple, overlapping taxonomies of AI coexist without clear coordination. This conceptual slippage is particularly salient in discussions of democracy, where references to LLMs often function as a stand-in for user-facing chatbot interfaces, despite the fact that democratic interactions occur at the level of the interface or application, rather than through direct engagement with the underlying model. Such uses conflate distinct technical layers (models, systems, and interfaces), thereby obscuring the specific loci at which democratic effects, risks, or interventions are assumed to arise.

4.2 RQ1: What Notions of Democracy are Discussed?

4.2.1 Definitions of Democracy

Across the literature, we examined the extent to which authors explicitly define democracy, in order to understand which dimensions of democracy are placed in relation to AI and from which theoretical traditions these understandings are derived or mobilized. As with the concept of AI discussed above, definitional clarity proves limited: nearly half of the corpus (47%) does not define democracy (see Table 2). Of these, 16 papers (so almost 3/5 of the 47%) also did not define AI, totalling 28% of all 57 articles.

The threshold for inclusion in the “definition” category was intentionally permissive. Articles were counted as defining democracy not only when offering formal definitions *stricto sensu*, but also when describing or explaining democratic principles or characteristics. Even under this relatively low threshold, explicit conceptualization remains rare. As visualized in Table 2, we therefore further disaggregated the “definition” category. Only 12 of 57 papers (21%) explicitly specify what they mean by democracy. One article adopts a meta-conceptual approach, treating democracy as a multifaceted phenomenon without committing to a single model (Jungherr, 2023). Seven papers (12%) provide implicit definitions by situating their arguments

Table 2 Definitions of democracy in the reviewed literature, categorised by degree of conceptual explicitness

Defined	Explicit definition	12
	Meta-conceptual engagement	1
	Implicit definition	7
	Underspecified definition	10
Undefined	No definition	27

optimizing bundles of values while not engaging political theory directly. As a consequence, democracy frequently appears less as a substantive or thick political concept and more as a checklist against which technological systems are evaluated.

4.2.2 Which notions of democracy does AI threaten or support?

Conceptions of democracy frequently overlap in the literature. Many contributions, for example, combine representative democracy with deliberative ambitions, while liberal democracies are often also representative in institutional form. For this reason, the categories should not be understood as mutually exclusive democratic models but as analytical classifications reflecting the primary democratic concern emphasized by a given contribution. Contributions were categorized according to the conception of democracy most explicitly mobilized by the authors. In practice, articles focusing primarily on rights and constitutional protections were classified as liberal democratic, whereas articles concerned with political representation and representative institutions were classified as representative democratic.

Against this backdrop, an important pattern emerges: a large share of papers that mobilise a form of deliberative democratic frameworks identify both significant opportunities for AI to support democratic practices, while also displaying the most variation of stances (see Fig. 3). On the supportive side, some authors highlight the productive role of large language models in policymaking (Špecián, 2024), the potential for democratic empowerment through chatbots (Croce, 2025), the importance of deliberative design in LLMs (Delacroix, 2025), and the use of AI for policy translation and inclusion, particularly in linguistically diverse contexts (Zidouemba, 2025). Others suggest that AI may help moderate or de-escalate polarized public debates (Cavaliere & Romeo, 2022). At the same time, concerns are raised that AI systems may undermine core deliberative norms such as reason-giving when acting as decision-makers (Beckman et al., 2024) and that chatbot-mediated communication may fragment rather than unify the public sphere (Cupać et al., 2024). Thus, deliberative democracy emerges as a domain of both high expectation and heightened concern.

A similar pattern is evident in work emphasizing representative democracy. Here, AI is often framed as a tool that could strengthen the relationship between representatives and their constituencies; for example, by functioning as an advisory system that helps representatives better align policy positions with constituent preferences (Lim & Savulescu, 2025), by using AI to support representation through liquid democracy (Sala, 2022), to represent new actors such as ecological systems (Bourgeois-Gironde, 2025), and to aggregate preferences in ways that align policy more closely with citizen preferences (Diallo et al., 2024).

By contrast, the participatory democratic strand might be expected to exhibit stronger scepticism toward AI (as AI may replace human participation), yet it registers comparatively low levels of outright threat ($n=1$). Instead, it has $n=3$ articles that stand in the category of ‘unclear’, meaning that they did not conclude in whether AI threatens or supports democracy (or both), and had $n=2$ categorised as ‘agnostic’, articles that provide a framework for assessment instead of developing a specific stance. The low level of ‘threat’ in those articles developing a participatory understanding of democracy may reflect the current undecidedness or tension between AI’s

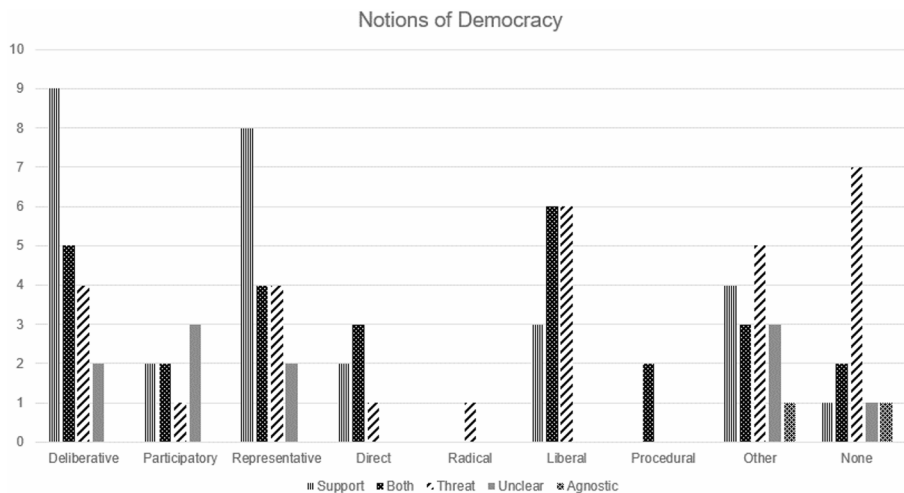


Fig. 3 Assessments of AI's democratic implications across different democratic notions

potential to facilitate participation and the concern that AI acting as a stand-in for citizens violates the core participatory requirement of direct human involvement in politics. Cohen and Suzor (2024), for example, argue that participation must be safeguarded through rights such as contestability (Contestable AI) rather than delegated to automated systems, while Delacroix (2025) argues that LLMs may, although are currently not, designed to enhance democratic participation.

Liberal democratic perspectives are, on a relative scale, the least favourable toward AI. Within this strand, AI is frequently framed as a manipulative force: whether through disinformation that threatens electoral integrity (Nasi, 2025), dynamics that facilitate democratic backsliding toward illiberalism (Fink-Hafner, 2025), the replacement of voting through predictive systems (Geddes, 2024), or the expansion of surveillance practices that undermine civil liberties and political equality (Gentzel, 2021). This contrasts with König and Wenzelburger's (2020) analysis of AI's implications on liberal democracy as they specifically investigate positive and negative impact scenarios.

Notably, papers that do not explicitly articulate a conception of democracy tend to exhibit the highest levels of perceived threat ($n=7$) and the fewest purely positive ($n=1$) and few exhibitions of ambivalence ($n=4$). By contrast, contributions grouped under "other" democratic notions, often more narrowly specified or domain-specific, are comparatively more optimistic ($n=7$). This may be because such accounts focus on concrete applications and localized risks as opposed to systemic or large-scale democratic threats.

Overall, the findings suggest that AI is neither uniformly framed as supportive of nor as threatening to democracy. The same technological features identified as enabling democratic support may simultaneously generate democratic risks, depending on the underlying conception of democracy employed. For example, AI systems that aggregate citizen preferences may be interpreted as enhancing responsiveness and representation, yet the same processes can be criticized for diminishing participa-

tory initiative or undermining citizens' autonomy. Similarly, large language models are described as supporting policymaking or facilitating policy translation and inclusion, while at the same time raising concerns about the erosion of deliberative norms such as human reason-giving and the potential fragmentation of the public sphere.

From this we see that some democratic notions are often mobilized to explore where AI might strengthen democracy, but this does not confirm a conclusion that specific conceptions of democracy are, on their own, optimistic or pessimistic about AI. What we can see is that researchers in our data tend to mobilize certain notions and examine how AI aligns. Thus they may for the sake of scholarly focus not concentrate on tensions elsewhere between AI and the same notion of democracy. In other words, AI may appear supportive from one analytical angle while simultaneously posing challenges to other core dimensions of the same democratic notion.

4.3 RQ2: Which Debates About Democracy are Raised?

Our data reveal six significant debates concerning democracy and AI. First, there is debate over the form of governance, specifically, who or what should govern (Sect. 4.3.1). Second, scholars disagree about who or what are, or should be, the agents of democracy (Sect. 4.3.2). Third, there is contention over whether AI should be regulated and thus restricted, or broadly implemented and actively promoted (Sect. 4.3.3). Fourth, debates persist over what “democratization” means in the context of AI (Sect. 4.3.4). Fifth, researchers examine whether, and in what ways, AI can strengthen or undermine one of the most heralded institutions of democratic systems: voting (Sect. 4.3.5). Finally, there is a growing body of scholarly work proposing different frameworks for understanding and assessing the relationship between democracy and AI (Sect. 4.3.6).

4.3.1 Forms of Government

The first group of debates concerns fundamental questions and can itself be divided into two strands. The first centres on the form of government, particularly debates that contrast machine-driven epistocracy (i.e., rule by the knowledgeable or politically competent) and/or technocracy (i.e., rule by technical and scientific experts) with popular, mass democracy. Some authors argue that AI-supported technocratic arrangements may improve policymaking overall (Hughes, 2017). Others, however, introduce important caveats, contending that such improvements are possible only if citizens retain control over societal direction and explicitly authorize AI involvement in legislative or decision-making processes. These arrangements are often described as weak forms of technocracy or “weak optipolitics” (Sætra, 2020; Coeckelbergh & Sætra, 2023; Stenseke, 2025; Slater, 2025).

Stronger forms of delegation, including AI rule or epistocracy, are rejected by many scholars, who emphasize that democratic authorization and collective political responsibility cannot be delegated to machines. Donahue (2025), for example, argues that advocates of epistocracy fail to account for the importance of group-level agency: citizens, understood as a collective, must be able to develop what he terms a “record of failure,” or *collective moral achievement*. On this view, forms of government in

which AI does not rule are normatively superior, because they alone preserve the conditions under which such collective responsibility can emerge. Similarly, Innerarity (2024) frames the debate around which form of government is most aligned with what is distinctively human in human agency, arguing that AI cannot, in principle, take over human governance. In line with this position, even scenarios of benevolent AI governance are criticized on the grounds that AI rule would undermine human agency and risk displacing politics itself (Damjanović, 2015; Kapelner, 2019).

Considered together, this strand of the literature uses AI as a philosophical probe or measuring rod to identify what is uniquely human about politics. Much like debates on the difference between artificial and human intelligence, these discussions raise a normative question: if humans were replaced in democratic decision-making, to what extent would the resulting system still qualify as a human democracy, and what would this imply for our concept of democracy itself? Considerable work remains to be done in this area.

Related contributions seek to articulate alternative pathways that integrate AI without abandoning democratic commitments. Simon (2022), for instance, proposes the concept of Linked Democracy, which explores how AI might be incorporated into democratic systems without compromising democratic principles. Drawing on the idea of militant democracy, Simon argues that democracies may need to take anti-democratic measures to protect themselves against corrosive forces such as authoritarianism and fascism that can otherwise flourish under unrestricted freedoms. Farrell (2025) similarly notes that political science has lagged behind other disciplines in assessing the democratic implications of AI. He distinguishes between AI as a *technology* of governance, used to assist with tasks such as information classification, and AI as a *form* of governance, which grants AI a foundational degree of autonomy and opens the possibility of novel, previously unseen forms of government.

A further debate under this heading concerns whether AI, if it is to contribute positively to democracy, necessarily shifts the form of government itself. Some argue, for example, that if AI were to resolve key limitations of representative democracy, democratic systems might move toward more direct forms of democracy by making voting mechanisms more flexible and responsive (Nasi, 2025). Others contend that liberalism is the dominant organizing principle of contemporary democracy, yet fundamentally incompatible with AI. From this perspective, social systems theory offers a more promising framework for accommodating AI. Within such a framework, AI cooperatives may address challenges posed by liberalism that liberal democratic theory itself struggles to resolve (Benthall & Goldenfein, 2021). In contrast, scholars such as St'ahel (2025) argue that constitutional or representative, or what he terms "normal", democracy is incapable of preventing the environmental devastation associated with AI-driven systems. On this view, prevailing democratic frameworks must be transformed into an "environmental democracy," one in which planetary thresholds cannot be exceeded and which is an issue that cannot be adequately regulated within existing democratic institutions.

Taken together, debates on forms of government reveal AI functioning as a normative testing ground for democracy itself. While limited forms of AI-assisted technocracy are sometimes considered compatible with democratic governance, stronger forms of delegation are largely rejected due to concerns about human agency, col-

lective responsibility, and democratic authorization. Rather than replacing democracy, much of the literature explores how AI might be integrated without displacing politics, prompting again reflection on what remains distinctively human about democratic rule. At the same time, some contributions suggest that AI may gradually reshape institutional arrangements, potentially enabling new democratic configurations or exposing limits within existing liberal and representative frameworks. Overall, this debate shifts attention from whether AI should govern to how democratic forms of government might evolve under conditions of increasing AI mediation.

4.3.2 Agents of Democracy

Alongside debates about the constitutional foundations of democracy, a growing body of work addresses questions of personhood and political inclusion, focusing on whether and how AI might be represented within democratic systems. Salmi (2023) explores the possibility of coexistence between humans and artificial general intelligences (AGIs) in democratic societies, treating AGIs as potential members of the political community. Beckman and Rosenberg (2022) similarly examine whether AI should be included in the *demos*, and on what normative grounds AI might retain agency or qualify as a political subject, including the possibility of legal personhood. Extending these arguments beyond AI itself, Bourgeois-Gironde (2025) proposes that AI could function as a representational vehicle for ecosystems and other non-conscious entities that may also be understood as legal persons. This approach, described as technological constitutionalism, conceptualizes AI agents as institutional intermediaries capable of representing non-human interests within democratic governance.

A related line of inquiry concerns whether AI, as a new agent within democratic systems, can contribute to democratic legitimacy. Cavaliere and Romeo (2022) argue that AI systems may enhance democratic legitimacy when societies face “rampant populism,” functioning as stabilizing and guiding instruments of governance, though only within frameworks that maximize political equality through the inclusion of diverse perspectives and robust oversight mechanisms. Gupta (2021), by contrast, focuses on the risks of AI interference, through the lens of a commodification of political influence. If electoral majorities can be assembled through “voter markets” analogous to commercial markets, then safeguarding procedural democracy, understood as the protection of minority interests and the maintenance of parity among competing interests, must be treated as a form of market regulation. This perspective raises fundamental questions about the meaning of voting and about how “the people” can sustain democratic authority without interference from non-political or commercially motivated actors.

Finally, several contributions examine the agential role of large language models in the epistemic dimensions of democratic policymaking. Coeckelbergh (2025) is sceptical of integrating LLMs into democratic decision-making, characterizing them as “bullshitters” whose outputs are agonistic with respect to truth and thus cannot function as an agent of democracy. More optimistic accounts, however, envision LLMs as advisory or enabling tools. Špecián (2024) argues that LLMs can serve a supportive role in policymaking; Croce (2025) highlights their potential to enhance democratic

access, deliberation, and representation by empowering citizens through chatbot-mediated interaction. Delacroix (2025) emphasizes that current chatbot designs risk undermining citizens' capacity to participate meaningfully in democracy, but does propose an alternative design framework in which LLMs function as "transitional spaces" for deliberation, potentially revitalizing democratic practice. Zidouemba (2025), finally, underscores the promise of AI in low- and middle-income countries, where LLMs may be used to translate public policies into official languages, thereby expanding democratic inclusion.

In sum, debates on agents of democracy expand the question of democratic participation beyond human actors, reconsidering who or what may legitimately count as part of the *demos*. The literature explores AI both as a potential political subject and as an institutional intermediary capable of representing human and non-human interests, thereby challenging established boundaries of political agency and personhood. While some scholars view AI systems and LLMs as tools that may enhance democratic legitimacy, inclusion, or epistemic support, others warn that automated agents risk distorting political authority, commodifying influence, or undermining truth-tracking deliberation. Across these contributions, AI emerges as a catalyst for rethinking democratic agency, representation, and the conditions under which political authority can be exercised.

4.3.3 Regulating or Implementing AI?

The literature on the governance of AI divides into two broad strands. The first emphasizes the regulation of AI and is oriented toward restriction. In this strand, AI is typically treated as an external force that must be constrained in order to function compatibly with democratic institutions or, at minimum, not interfere with them. At the philosophical level, Westerstrand (2024) argues that John Rawls's theory of justice as fairness can provide a normative foundation for AI principles and regulatory ethics guidelines that are otherwise philosophically and politically unsatisfactory. Without such justification, Westerstrand contends, AI principles risk amounting to little more than ungrounded opinion. In conjunction, Cupac et al. (2024) survey how the European Union has responded, through regulation, to harms that "impede democratic deliberation," identifying the instruments available to protect values such as transparency and digital literacy.

Several authors criticize the dominance of AI ethics as a governance paradigm. Bogiatzis-Gibbons (2024) argues for the need for both what he calls 'weak' regulatory frameworks like the GDPR and AI Act, but also calls for control in a 'stronger' political sense by affected communities of citizens. Carlsson and Rönnblom (2022) argue that the proliferation of ethics frameworks has displaced genuinely political regulation, allowing private-sector ethics committees to engage in self-regulation that effectively amounts to non-regulation. Gianni et al. (2022) similarly demonstrate that ethics frameworks often lack mechanisms of implementation and therefore cannot stand alone; instead, they must be substantiated, in their view, by a Deweyan conception of democratic politics. Erman and Furendal (2022) advance this critique of value embedding by rejecting what they call an "additive view" of legitimacy, where values are simply layered onto one another, and they argue instead in favour of

a holistic approach that integrates input legitimacy (i.e., the representation of affected members of society) and output legitimacy (i.e., the justification of binding collective decisions). Taken together, these contributions point to a partial but not yet fully articulated shift from ethics toward politics in debates about how AI regulation can preserve democratic integrity, across deliberative, liberal, and radical democratic traditions.

The second strand adopts a more suggestive orientation, focusing on how AI systems might be implemented in ways that actively support democratic practices. Cohen and Suzor (2024) argue that contestability plays a central role in democratically legitimizing AI systems, while Züger and Asghari (2023) similarly maintain that principles and procedures for “AI for Social Good” (AI4SG) can be developed in a democratically defensible manner. Ovadya (2023) likewise calls for a democratic adaptation of AI governance. From a more critical perspective on implementation, Paltieli (2022) analyses the goals of national AI implementation strategies through the lens of “political imaginaries” (e.g., data and consent), and concludes that the simultaneous commitment to both human-centeredness and the common good renders many national strategies “theoretically incoherent”.

Within this implementation-oriented literature, proposals range from normative refinements to evaluative systems. Benton (2025) argues that public reason, not only rights such as explainability or contestability, should play a legitimizing role for AI in democratic societies. Ovadya et al. (2025) propose a “Democracy Level Framework” for assessing the democratic character of AI systems by examining both the degree of automated decision-making and the meta-level capacity of systems to modify their own rules, thereby distinguishing genuinely democratic systems from those that merely simulate democratic values. Starting from the premise that democracy should be epistemic rather than primarily moral, Lim and Savulescu (2025) suggest that politicians could use AI to assess whether policy proposals align with constituency preferences, offering a framework that delineates acceptable uses and limits of AI in representative democracies. Likewise, Kim (2024) argues that “political AI” may be able to better reflect citizen needs. Han (2025), similarly concerned with strengthening democratic infrastructure, provides a matrix for different types of AI involvement in voting, arguing that while some configurations are undesirable, others may enhance democratic processes.

A subset of this literature is more explicitly optimistic about AI’s democratic potential. Burgess (2022), for example, explores algorithmic augmentation of democracy through thought experiments, arguing that “even the most extreme innovations” may help promote “basic principles that we may fundamentally value” and therefore “merit both continued and serious consideration.” Other authors, however, emphasize the democratic risks associated with AI implementation. Gentzel (2021) argues that bias in AI-based facial recognition technologies is incompatible with the democratic principle of equality before the law. The analysis rejects abandoning the equality ideal and instead calls for eliminating bias, highlighting how facial recognition exacerbates racial discrimination and undermines liberal democracy, even when discriminatory outcomes are not intentionally produced by system users. Black (2025) similarly warns that AI-driven surveillance threatens democracy by exposing citizens to politically influential content and by exacerbating inequalities in free time and

epistemic resources, arguing that surveillance capitalism disrupts democratic equality at a structural level.

Concerns about democratic agency are also prominent in governance debates over AI-driven content moderation on digital platforms. De Gregorio (2020) argues that the liberal tradition of *ex ante* regulation through terms of use has increasingly been reframed, particularly in the European Union, as a matter of rights. From this perspective, opaque or unaccountable AI-based moderation practices threaten the democratic right to free speech, while existing regulatory frameworks remain underdeveloped.

What emerges from these debates is a tension between restrictive and enabling approaches to AI in democratic contexts. While regulatory approaches increasingly emphasize political legitimacy over ethical guidelines, implementation-oriented approaches range from cautious, conditional proposals to more ambitious visions of democratic enhancement. Across both strands, a growing body of work seeks not only to regulate or deploy AI, but also to develop criteria for assessing whether, and under what conditions, AI can promote democracy.

4.3.4 Democratization of AI

Several contributions address the democratization of AI in ways that go beyond a narrow, *prima facie* understanding of democratization as mere availability or access. Himmelreich (2023) argues that the central question is not how AI can be democratized in the sense of wider access, but how “the organizational forms and modes of governance” shaping AI development and deployment can themselves be made democratic. On this view, democratization concerns how AI is implemented within systems of governance, not simply who can use AI tools.

Other studies emphasize the multifaceted character of democratization. Lin (2024) argues that successful efforts to democratize AI can counteract algorithmic injustice, and distinguishes between three dimensions of AI democratization: democratizing AI use, democratizing AI development, and democratizing AI governance. The first concerns access beyond technical specialists; the second involves lowering barriers to participation in AI development by expanding opportunities for involvement; and the third entails subjecting AI governance to collective decision-making processes. Rubeis et al. (2022) further demonstrate that even within a specific domain, such as healthcare, the notion of democratization is politically laden and refers to different possible types of *demos*: “democratization” may variously designate biomedical professionals, patients, technology-enabled users, or society at large as the relevant political subject. This plurality of referents underscores that democratization is not a uniform or self-evident concept, but one that is contested and context-dependent, even within narrowly defined policy domains.

4.3.5 Voting

A substantial debate concerns whether AI can enhance democracy by optimizing voting and preference aggregation in different ways. Scholars who adopt a more optimistic stance argue that machine-learning driven polling and prediction systems can generate highly accurate representations of public preferences, enabling policymak-

ers to govern more closely in line with the *demos* (Cerina & Rouméas, 2025; Lim & Savulescu, 2025). From the policymaker's perspective, AI may also function as an advisory tool for aligning political decisions with constituency preferences (Špecián, 2024). On this view, AI enables a new democratic model in which systems aggregate and process voter preferences and subsequently propose policies, identify compromises, or even govern under specified constraints (Diallo et al., 2024).

Relatedly, Sala (2022) argues that Liquid Democracy, in which voters may freely delegate their votes to others they deem better qualified, while retaining the ability to revoke that delegation at any time, is particularly compatible with AI technologies and addresses several legitimacy problems associated with contemporary voting regimes. Within this literature, *algorithmic democracy* acquires a meaning distinct from *algocracy* or technocracy. Whereas the latter denotes rule by experts (or epistemic AIs), algorithmic democracy, especially in the context of transformative voting systems, emphasizes responsiveness to citizens' preferences and proximity to the members of society. Algorithmic democracy and algocracy, as understood above, are not mutually exclusive, however, and may be combined in systems that implement policies "from above" on the basis of data aggregated "from below". Such hybrid arrangements raise concerns about whether aggregated data genuinely reflect the will of the *demos*, particularly in the absence of prior public deliberation, and remain an important subject for further research.

Other scholars are more sceptical, viewing AI as a threat to democratic voting. A recurring concern is manipulation, with AI framed as an external force capable of subverting liberal democracies by influencing voter behaviour and facilitating democratic backsliding (Fink-Hafner, 2025). Geddes (2024) examines whether AI-based vote prediction could legitimately replace voting altogether but concludes that such substitution would violate democratic autonomy. Similarly, Slater (2025) argues that neither representative nor direct democracy can be replaced by lottocracy (i.e., the idea that political offices are occupied by sortition rather than decided through elections) and, by extension, have AI make policy-decisions, since democratic *authorization* by members of society plays an indispensable role that cannot be delegated, neither to chance nor AIs. Even where AI is used, Slater maintains, human authorization processes must remain in place to approve or reject AI-supported decisions.

A third group of scholars emphasizes both the democratic risks and the potential benefits of AI in voting systems. Nasi (2025), for example, argues that while AI can degrade the public sphere through the proliferation of credible misinformation, it may also enhance representative democracy by increasing flexibility and responsiveness. If such enhancements are successful, Nasi suggests, they could ultimately transform representative democracy into more direct forms of democracy. Gupta (2021) leaves open the question of how "the people" can be meaningfully constituted in democratic systems when voters can be swayed through AI-driven manipulation, thereby connecting voting to broader concerns about the marketization of political influence. Casares (2018) similarly warns that democracy is threatened by cognitive machines, while nevertheless acknowledging domains, such as judicial decision-making crucial for democracy's rule of law, where eliminating certain forms of human error or bias through AI is normatively desirable.

4.3.6 Frameworks for Democracy

Across the literature, a distinct strand of scholarship addresses not the substantive effects of AI on democracy, but the question of how such effects should be evaluated in the first place. Instead of advancing binary claims about whether AI strengthens or undermines democratic governance, several authors develop analytical frameworks intended to structure assessment and comparison. These contributions propose criteria, dimensions, or scales through which democratic implications can be systematically examined: for example, frameworks for assessing democratic threats (Jungheer, 2023), normative desiderata for democratic and fair global AI governance (Erman & Furendal, 2022), evaluative models of democratic AI systems and automated decision-making (Ovadya et al., 2025), distinctions between acceptable and impermissible democratic uses of AI in representing public preferences (Lim & Savulescu, 2025), contextual approaches emphasizing the situated political nature of AI practices (Noorman & Swierstra, 2023), and graded models of political tasks according to their suitability for AI involvement (König & Wenzelburger, 2022). In sum, these works reflect a broader trend toward meta-level conceptualization, in which scholars seek to establish evaluative lenses of the relation between democracy and AI.

5 Discussion

This review set out to examine how democracy is conceptualized in research that explicitly relates AI to democracy. The findings show that democracy does not function as a stable or shared evaluative benchmark in this literature. Rather, it operates as a plural and shifting normative reference point. As a result, assessments of whether AI threatens or supports democracy (or both in different ways) frequently diverge because authors rely on different notions of democracy. What appears as contradiction at the level of conclusions thus often reflects deeper normative divergence at the conceptual level.

A second key finding concerns the way democratic theory is mobilized in relation to AI. The literature predominantly exhibits an “inheritance” structure, in which established models of democracy, most notably deliberative and representative democracy, are imported as evaluative frameworks. While this approach allows scholars to assess AI against familiar democratic standards, it rarely involves rethinking those standards in light of AI’s distinctive capacities and modes of intervention, although occasionally, specific premises or conceptions of democracy are broadened, expanded, or revised. Democracy is thus more often treated as a background concept that evaluates AI, and not as a political form whose conditions of agency, legitimacy, and authority may themselves be reshaped by AI. As a consequence, much of the literature remains classificatory rather than generative, applying democratic labels without systematically interrogating how AI might reshape the underlying foundations of democratic theory (form of government, agency, representation, personhood). However, a smaller, though significant, body of work *does* adopt a more generative approach, introducing concepts such as algocracy, optipolitics, technological constitutionalism and environmental democracy.

The review also reveals a notable relationship between conceptual explicitness and evaluative orientation. Contributions that do not explicitly work with a specific notion of democracy are more likely to frame AI as a significant threat to democracy. By contrast, studies that provide explicit notions of democracy tend to adopt more positive or conditional positions, often identifying both democratic risks and opportunities associated with AI. This pattern suggests that having no notion of democracy may amplify pessimism about AI's influence on democracy, whereas having a specific notion enables more analytically restrained and differentiated assessments. Importantly, this is not a claim about the normative correctness of optimism or pessimism, but about the role that notional specification plays in structuring the debates.

Finally, these findings have implications for how the current state of AI and democracy research should be understood. This review does not aim to adjudicate between competing democratic models, nor to determine whether AI is ultimately beneficial or harmful to democracy in general. Its contribution lies in mapping the conceptual landscape within which such judgments are made and in showing how democratic assumptions shape evaluations of AI's democratic implications. At the same time, the review underscores the limits of existing scholarship: without greater conceptual clarity regarding democracy, empirical findings and normative arguments risk talking past one another. In our view, the results suggest that progress in research on democracy and AI depends not only on further empirical investigation or technical refinement, but also on sustained conceptual work that makes democratic assumptions explicit and thus open to scrutiny.

6 Limitations

This review has several limitations that should be acknowledged. First, AI is a relatively recent and specific term from the 1950s (Copeland, 2023), and by using *AI* and *"artificial intelligence"* as core search terms, we may have excluded scholarship that addresses similar phenomena under different labels, such as machine intelligence or algorithmic reasoning. This creates a risk of missing relevant literature that develops or employs computational methods as standalone search terms that belong under the AI umbrella but without using the label. Our goal was to investigate the sociotechnical imaginary of AI in relation to democracy, as described above (Sect. 2), but the more granular labels at other levels of abstraction do persist as a body of relevant literature that may have not been fetched. However, including literature using adjacent terminology would arguably have introduced subject dispersion instead of a more suitable coverage.

Second, we deliberately excluded large bodies of literature on algorithms and social media, if they did not either directly or comprehensively relate to democracy and AI. While these literatures are clearly relevant to democratic theory and practice, they constitute a well-established and expansive research field in their own right. Including them would have significantly broadened the scope of the reviewed literature beyond feasibility and likely also blurred its analytical focus. Instead, we treat this literature as complementary.

Third, this review prioritizes peer-reviewed journal articles and conference proceedings and therefore excludes preprint repositories and working-paper platforms such as arXiv and SSRN. This decision reflects a trade-off between inclusiveness and methodological clarity. While these repositories contain influential and timely work, systematically incorporating them would raise demarcation challenges regarding quality control and boundary-setting. As a result, some relevant but non-peer-reviewed contributions may be absent from our analysis.

Finally, our methodological choices may have shaped which conceptions of democracy are represented in the reviewed literature. Our approach favours studies that explicitly define, theorize, or problematize democracy in relation to AI, and this has led to the exclusion of work that relies on a fixed or “taken-for-granted” understanding of democracy, for example, studies that operationalise a specific democratic model within an application or system without engaging in conceptual discussion. Consequently, the resulting corpus may reflect biases in how our findings represent notions of democracy.

These limitations highlight that this review provides a structured mapping of a particular slice of the literature, not an exhaustive account of all work relevant to democracy and AI.

7 Future Research

We will highlight three directions for future research on democracy and AI.

First, there is a need for greater conceptual precision in how democracy is defined and operationalised in AI-related scholarship. Instead of relying on implicit assumptions or broad value lists, future work should treat the specification of democracy as a methodological advantage. Explicit definitions, whether substantive, pluralist, or meta-theoretical, would enable clearer comparison across studies and reduce the tendency toward unproductive disagreement driven by unarticulated normative commitments.

Second, future research would arguably benefit from moving beyond the application of established democratic models toward more generative engagements between democratic theory and AI. Much of the existing literature evaluates AI against inherited conceptions of liberal, deliberative, or representative democracy, while leaving these frameworks largely unchanged, at least insofar as their theoretical articulation within the relevant articles can be assessed. Yet as AI systems may increasingly mediate political representation, decision-making, and the exercise of epistemic authority in ways that may challenge foundational democratic assumptions, developing democratic theory with AI, rather than merely applying it to AI and discussing it, represents an important agenda for political theory, philosophy of technology, and related fields.

A further direction for future research concerns the limited presence of feminist, postcolonial, and Global South perspectives within the AI–democracy literature. With few exceptions, such as isolated engagements with environmental democracy

or low- and middle-income contexts, the corpus is largely shaped by democratic frameworks rooted in Western liberal and deliberative traditions. To be sure, this does not mean that deliberative or representative models are incompatible with feminist or decolonial approaches; these traditions have long been critically reworked from such perspectives. However, their relative absence in current scholarship arguably narrows the normative scope of debate, and future research could more systematically explore them.

8 Conclusion

This review has examined how democracy is conceptualized in scholarship on AI and how these conceptualizations shape assessments of AI's implications for democratic governance in the literature. The results show that democracy functions less as a stable evaluative benchmark than as a plural and often under-specified normative reference point. Different democratic traditions, such as liberal, deliberative, representative, and participatory models, are invoked unevenly and often without sustained theoretical elaboration. Assessments of AI's democratic implications are structured by divergent and sometimes implicit assumptions about what democracy entails.

From this, the review maps three key dynamics currently at play in the literature. First, conceptualizations matter: Evaluations of AI's democratic implications are determined by the democratic frameworks through which it is interpreted. Different conceptions of democracy generate different assessments of whether AI strengthens or undermines democratic governance. Second, little conceptual reworking: The review identifies a recurring "inheritance structure" in the literature, whereby established democratic models are applied to AI systems or governance arrangements without substantial reconsideration of how AI may transform the underlying conditions of democracy. Third, clarity produces nuance: Conceptual clarity plays a decisive role in shaping normative assessments. Studies that articulate explicit democratic frameworks tend to produce more differentiated and ambivalent evaluations of AI, whereas studies relying on implicit or underspecified notions of democracy more often portray AI as an overarching democratic threat.

Taken together, these mapped dynamics suggest that research on democracy and AI faces a conceptual challenge as much as an empirical or technical one. Debates about AI's democratic implications are ultimately debates about how democracy itself is understood. Without greater clarity about the democratic models (and their premises and specificity) being invoked, empirical findings and normative arguments risk talking past one another. AI should therefore not only be evaluated against established democratic understandings but also understood as a development that invites renewed reflection on the assumptions, boundaries, and transformations of democratic theory. Making these assumptions explicit is therefore a necessary step toward a theoretically grounded research agenda on AI and democracy.

Appendix

Table 4 Overview of included papers in the review including author codings

Paper	Notion of democracy	Type of AI	Un-defined AI	Does AI support or threaten democracy?	Definition of democracy category
Beckman and Rosenberg (2022)	None	Narrow AI; Artificial General Intelligence	X	Agnostic	No definition
Beckman et al. (2024)	Deliberative	Machine Learning		Threaten	Underspecified definition
Benthall Goldenfein (2021)	Liberal		X	Threaten	No definition
Benton (2025)	Deliberative; Liberal; Other		X	Both	No definition
Black (2025)	None		X	Threaten	No definition
Bogiatzis-Gibbons (2024)	Deliberative; Participatory	Other	X	Both	Implicit definition
Bourgeois-Gironde (2025)	Representative		X	Support	No definition
Burgess (2022)	Representative; Direct; Other	Other	X	Both	Explicit definition
Carlsson and Rönblom (2022)	Radical; Liberal; Other		X	Threaten	Explicit definition
Casares (2018)	None		X	Threaten	Underspecified definition
Cavaliere and Romeo (2022)	Deliberative; Other			Support	Explicit definition
Cerina and Rouméis (2025)	None	Machine Learning; Narrow AI; Other		Support	Underspecified definition
Coeckelbergh (2025)	Deliberative; Participatory; Other	Large Language Models		Threaten	No definition
Coeckelbergh and Sætra (2023)	Deliberative		X	Both	No definition
Cohen and Suzor (2024)	Participatory		X	Unclear	No definition
Croce (2025)	Deliberative; Representative	Large Language Models		Support	No definition
Cupać et al. (2024)	Deliberative			Threaten	Explicit definition
Damnjanović (2015)	Other	Superintelligence		Threaten	Implicit definition
DeGregorio (2020)	Liberal	Machine Learning	X	Threaten	No definition
Delacroix (2025)	Deliberative; Participatory	Large Language Models		Support	Underspecified definition
Diallo et al. (2024)	Deliberative; Participatory; Representative; Direct; Other	Large Language Models		Support	No definition
Donahue (2025)	None		X	Threaten	Underspecified definition
Erman and Furendal (2022)	Other	Superintelligence; Artificial General Intelligence	X	Both	Explicit definition

Table 4 (continued)

Paper	Notion of democracy	Type of AI	Un-defined AI	Does AI support or threaten democracy?	Definition of democracy category
Farrell (2025)	None	Large Language Models; Machine Learning	X	Both	No definition
Fink-Hafner (2025)	Representative; Liberal; Other	Large Language Models; Machine Learning; Other		Threaten	Explicit definition
Geddes (2024)	Deliberative; Representative; Liberal			Threaten	Implicit definition
Gentzel (2021)	Liberal	Other		Threaten	No definition
Gianni et al. (2022)	Other		X	Both	Implicit definition
Gupta (2021)	Procedural	Other		Both	Explicit definition
Han (2025)	Deliberative	Machine Learning	X	Support	No definition
Himmelreich (2023)	Deliberative; Participatory; Other	Machine Learning	X	Unclear	Explicit definition
Hughes (2017)	Other	Superintelligence; Other		Support	No definition
Innerarity (2024)	None		X	Threaten	Explicit definition
Jungherr (2023)	Other	Large Language Models; Narrow AI; Artificial General Intelligence; Deep Learning		Threaten	Meta-conceptual engagement
Kapelner (2019)	None	Superintelligence; Artificial General Intelligence		Threaten	No definition
Kim (2025)	Other	Artificial General Intelligence	X	Support	Implicit definition
König and Wenzelburger (2020)	Liberal	Machine Learning; Deep Learning		Both	Explicit definition
König and Wenzelburger (2022)	Procedural; Liberal; Other	Machine Learning; Narrow AI		Both	Underspecified definition
Lim and Savulescu (2025)	Deliberative; Representative; Liberal	Large Language Models; Other		Support	Implicit definition
Lin (2024)	Deliberative; Representative; Direct			Both	No definition
Nasi (2025)	Representative; Direct; Liberal; Other	Other		Both	Underspecified definition
Noorman and Swierstra (2023)	Other		X	Agnostic	Explicit definition
Ovadya (2023)	Deliberative; Representative; Other		X	Support	No definition
Ovadya et al. (2025)	Other		X	Both	Underspecified definition
Paltieli (2022)	Liberal	Narrow AI		Support	No definition

Table 4 (continued)

Paper	Notion of democracy	Type of AI	Un-defined AI	Does AI support or threaten democracy?	Definition of democracy category
Rubeis et al. (2022)	Deliberative; Participatory; Representative; Other	Machine Learning	X	Unclear	No definition
Sala (2022)	Deliberative; Representative; Direct; Other		X	Support	No definition
Salmi (2023)	Representative	Superintelligence		Both	No definition
Simon (2022)	Deliberative; Liberal; Other		X	Both	Underspecified definition
Slater (2025)	Representative; Direct	Other	X	Threaten	No definition
Špecián (2024)	Representative; Other	Large Language Models		Support	No definition
St'ahel (2025)	Representative; Other			Threaten	Explicit definition
Stenseke (2025)	Representative; Liberal	Superintelligence; Other		Support	No definition
Sætra (2020)	None	Machine Learning; Deep Learning; Other	X	Both	No definition
Westerstrand (2024)	Liberal; Other	Large Language Models; Other	X	Both	Implicit definition
Zidouemba (2025)	Participatory	Other		Both	Underspecified definition
Züger et al. (2023)	Deliberative		X	Support	No definition

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